# BOUNDARY FIBERS IN FREE DE MORGAN ALGEBRAS: BOOLEAN INTERVALS, SPERNER BOUNDS, AND MEDIAN CONTRACTIONS

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Abstract. In the free De Morgan algebra on $n$ generators, the *boundary* of an element is the tuple of its $2n$ face restrictions, obtained by substituting the endpoints $0, 1$ for a generator. We determine the fibers of the boundary map completely: each fiber is an interval which is a Boolean lattice $2^D$, where $D$ is an antichain of *diagonal* points of the underlying literal cube; conversely every antichain of diagonal points arises, so by Sperner's theorem the largest fiber has exactly $2^{\binom{n}{\lfloor n/2 \rfloor}}$ elements. Fibers are connected by an explicit median-style interpolation with one fresh generator, constant on all faces — a strict cylinder. We further show that every compatible sphere of faces is the boundary of some element, and that boundaries are in bijection with antichains of *non*-diagonal points, which reduces the enumeration of boundaries to a Dedekind-type antichain count. A complete machine census for $n \leq 3$ (all 7,828,354 elements at $n = 3$) accompanies the paper; none of the resulting sequences appear in the OEIS. The questions are motivated by the interval algebras of cubical type theory, where the fibers describe the strict (judgmental) fillers of a fixed boundary.

## 1. Introduction

A De Morgan algebra is a bounded distributive lattice equipped with an involution $\neg$ interchanging $\wedge, \vee$ and $0, 1$; unlike a Boolean algebra it does not satisfy the law of the excluded middle, so $x \wedge \neg x$ need not vanish. The finitely generated free De Morgan algebras $\mathrm{DM}_n$ are classical objects [1, 6, 7, 8]: as a lattice, $\mathrm{DM}_n$ is the free bounded distributive lattice on the $2n$ *literals* $x_1, \neg x_1, \ldots, x_n, \neg x_n$, hence is isomorphic to the lattice of monotone Boolean functions of $2n$ variables, of cardinality the Dedekind number $M(2n)$.

This paper studies a natural map on $\mathrm{DM}_n$ that appears not to have received attention in the lattice-theoretic literature: the *boundary map*. Substituting an endpoint $e \in \{0, 1\}$ for a generator $x_i$ (hence $1 - e$ for $\neg x_i$) defines a surjective homomorphism $\partial_i^e : \mathrm{DM}_n \to \mathrm{DM}_{n-1}$, the *face* maps; the boundary of $\varphi \in \mathrm{DM}_n$ is the tuple $\partial\varphi = (\partial_i^e \varphi)_{i \leq n,\, e \in \{0,1\}}$ of all its faces.

2020 *Mathematics Subject Classification.* 06D30; 06A07, 05A15, 03B38.

*Key words and phrases.* De Morgan algebra, free algebra, distributive lattice, monotone Boolean function, antichain, Sperner's theorem, median, Dedekind number, cubical type theory.

Two questions arise at once. *Which* boundaries occur? And what is the structure of the *fiber* — the set of all elements with a given boundary?

Our main theorem answers the second question completely. Call a point of the literal cube $\{0,1\}^{2n}$ *diagonal* if, for every $i$, its pair of coordinates for $(x_i, \neg x_i)$ is $(0,0)$ or $(1,1)$ — the points invisible to every face.

**Theorem** (Theorem 3.4)**.** *Every fiber of the boundary map on* $\mathrm{DM}_n$ *is an interval* $[m, M]$ *whose difference* $D = M \setminus m$ *(in the monotone-function representation) is an antichain of diagonal points, and the fiber is a Boolean lattice, canonically isomorphic to* $2^D$*. Conversely, every antichain of diagonal points arises as the* $D$ *of some fiber. Consequently the maximal fiber cardinality in* $\mathrm{DM}_n$ *is exactly* $2^{\binom{n}{\lfloor n/2 \rfloor}}$.

The appearance of Sperner's theorem [14] is, to our knowledge, new in this context: the diagonal points form a copy of the Boolean cube $\{0,1\}^n$, and the extremal fibers are governed by its maximal antichains. The proof is short and entirely elementary — the crux is that between any two comparable diagonal points there lies a face point (Lemma 3.3) — but the statement is sharp and, as the machine census shows, the Boolean structure is far from sparse: at $n = 3$ there are 2,130,934 non-singleton fibers, and the explicit complement $g \mapsto m \vee (M \wedge g^c)$ witnesses Boolean-ness in every one of them.

Around the main theorem we prove three companion results. First, a *filling theorem* (Theorem 4.2): every compatible sphere — a tuple of prospective faces satisfying the evident pairwise conditions — is the boundary of an element, with a canonical minimal filler. Second, a *median contraction* (Theorem 5.1): for $\varphi, \varphi'$ in one fiber, the element

$$\chi \;=\; (\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi') \;\in\; \mathrm{DM}_{n+1}$$

in one fresh generator $k$ restricts to $\varphi$ at $k = 0$, to $\varphi'$ at $k = 1$, and is *constant in* $k$ *on every face* — a cylinder connecting the two fillers relative to their common boundary, inside the free algebra, with no completeness assumptions. The ternary shape is the classical median of Birkhoff–Kiss [3] adapted to the involutive setting; the fibers are in particular median-closed subsets in the sense of median-algebra theory [2]. Third, a *classification of boundaries* (Theorem 4.3): realizable boundaries are in bijection with antichains of *non-diagonal* points of the literal cube, via the minimal filler; this reduces the first question above to a Dedekind-type antichain enumeration.

Section 6 reports a complete computational census for $n \leq 3$ and a targeted construction at $n = 4$. Writing $B(n)$ for the number of realizable boundaries:

$$B(1) = 4, \qquad B(2) = 82, \qquad B(3) = 4{,}829{,}136,$$

with fiber-size distributions $\{1^2, 2^2\}$, $\{1^{28}, 2^{38}, 4^{16}\}$ and

$$\{1^{2698202},\ 2^{1747364},\ 4^{358284},\ 8^{25286}\}$$

respectively — all sizes powers of two, as the main theorem predicts, with the counting identity $M(2n) = \sum_{\text{fibers}} 2^{|D|}$. None of these sequences occur in the OEIS at the time of writing.

Motivation from cubical type theory. The free De Morgan algebra is the interval theory of cubical type theory in the style of Cohen–Coquand–Huber–Mörtberg [4]: an $n$-cube context has its cells reparametrized along elements of $\mathrm{DM}_n$, and the equations that hold in $\mathrm{DM}_n$ are exactly the equations the proof assistant's kernel decides *judgmentally.* The fiber of the boundary map then describes all *strict* fillers of a fixed cubical boundary — the fillers available before any Kan composition is invoked. Non-uniqueness of strict fillers, the median interpolation, and the strict contractibility that Theorem 3.4 refines were first observed, machine-checked, in a proof-assistant kernel built by the author [12, 13]; the present paper isolates the underlying lattice theory, which is independent of type theory and, we believe, of interest in its own right. In the recent literature, Doré–Cavallo–Mörtberg [5] automate the *search* for a filler of a given boundary over the same algebras; their representation of candidate solutions is deliberately lossy about the solution *set*, and they isolate — in two sentences of prose on "consistent" assignments [5, §4.3] — the combinatorial subtlety specific to the De Morgan case, namely that boundaries constrain only part of the literal cube. The present results resolve that subtlety into a complete structure theorem for the solution set: the unconstrained diagonal points are exactly the free Boolean degrees of freedom of Theorem 3.4. Enumerative prior art on interval sizes in the monotone-function lattice is due to Pawelski [11]; our fibers are particular intervals selected by the boundary conditions, and our census concerns their distribution.

## 2. Preliminaries

2.1. **Free De Morgan algebras as monotone functions.** A *De Morgan algebra* $(A, \wedge, \vee, 0, 1, \neg)$ is a bounded distributive lattice with a unary $\neg$ satisfying $\neg\neg a = a$ and $\neg(a \wedge b) = \neg a \vee \neg b$. We write $\mathrm{DM}_n$ for the free De Morgan algebra on generators $x_1, \ldots, x_n$. The following classical description is the combinatorial engine of everything below.

The description below is stated in [1, Ch. XI] and [6], and in sharpest form in [8, Thm. 3.2, Cor. 2.5] (cf. [7]).

Let $Q_n := \{0,1\}^{2n}$ be the *literal cube*, with coordinates grouped in pairs; a point is $\alpha = (\alpha_1, \ldots, \alpha_n)$ with $\alpha_i \in \{0,1\}^2$, the $i$-th pair recording prospective values of $(x_i, \neg x_i)$ *independently.* Order $Q_n$ componentwise. Then $\mathrm{DM}_n$ is isomorphic, as a bounded lattice, to the lattice of monotone functions $Q_n \to \{0,1\}$, equivalently of *up-sets* of $Q_n$, with the generators represented by the coordinate projections; the involution is $(\neg\varphi)(\alpha) = 1 - \varphi(\sigma\bar{\alpha})$, where $\bar{\alpha}$ complements all coordinates and $\sigma$ swaps each pair. In particular $|\mathrm{DM}_n| = M(2n)$, the Dedekind number of monotone functions on $2n$ variables [10, 5]. We identify elements with their up-sets and write $\varphi \subseteq \psi$ for the lattice order, $\varphi^c$ for the *pointwise* complement of the up-set (a down-set, in general not an element of $\mathrm{DM}_n$).

2.2. **Faces, boundaries, and the two kinds of points.** For $i \leq n$ and $e \in \{0,1\}$ the *face map* $\partial_i^e : \mathrm{DM}_n \to \mathrm{DM}_{n-1}$ substitutes $x_i := e$, i.e. fixes the $i$-th pair to $(0,1)$ if $e = 0$ and to $(1,0)$ if $e = 1$, and forgets it. Face maps are surjective lattice homomorphisms and satisfy the evident cubical identities $\partial_j^\delta \partial_i^\varepsilon = \partial_i^\varepsilon \partial_j^\delta$ (indices adjusted). The *boundary* of $\varphi$ is $\partial\varphi := (\partial_i^e \varphi)_{i,e}$, and the *fiber* of a boundary is the set of all $\psi$ with $\partial\psi = \partial\varphi$.

Call $\alpha \in Q_n$ a *face point* if some pair $\alpha_i \in \{(0,1),(1,0)\}$, and a *diagonal point* if every pair $\alpha_i \in \{(0,0),(1,1)\}$. Face points are exactly the points lying in some face subcube — the points that some face map sees — while the $2^n$ diagonal points form a copy of the Boolean cube $\{0,1\}^n$ (send $(1,1) \mapsto 1$, $(0,0) \mapsto 0$), invisible to every face. This dichotomy, specific to the paired (involutive) situation, is what makes the De Morgan case genuinely different from a free distributive lattice with unconstrained substitutions; cf. the discussion in [5].

**Lemma 2.1.** *If $\alpha < \beta$ are comparable diagonal points, there is a face point $\gamma$ with $\alpha < \gamma \leq \beta$.*

*Proof.* Some pair has $\alpha_i = (0,0)$ and $\beta_i = (1,1)$. Let $\gamma$ agree with $\alpha$ except $\gamma_i := (0,1)$. Then $\alpha < \gamma$, $\gamma \leq \beta$, and $\gamma$ is a face point. □

## 3. The structure theorem

**Lemma 3.1.** *Every fiber is an interval: it is closed under $\wedge, \vee$ (hence has a least element $m$ and greatest element $M$), and every $\psi$ with $m \subseteq \psi \subseteq M$ lies in the fiber.*

*Proof.* Faces are lattice maps, so fibers are sublattices; finiteness gives $m, M$. If $m \subseteq \psi \subseteq M$ then $\partial_i^e m \subseteq \partial_i^e \psi \subseteq \partial_i^e M$ by monotonicity of $\partial_i^e$, and the outer terms are equal. □

**Lemma 3.2.** *For a fiber $[m, M]$, the difference $D := M \setminus m$ contains no face point.*

*Proof.* A face point of $M \setminus m$ would make the corresponding faces of $M$ and $m$ differ. □

**Lemma 3.3.** *$D$ is an antichain.*

*Proof.* Suppose $\alpha < \beta$ in $D$; both are diagonal by Lemma 3.2. Pick $\gamma$ as in Lemma 2.1. Since $\alpha \in M$ and $M$ is an up-set, $\gamma \in M$; since $\gamma$ is a face point, $\gamma \notin D$, so $\gamma \in m$; since $m$ is an up-set and $\gamma \leq \beta$, $\beta \in m$, contradicting $\beta \in D$. □

**Theorem 3.4** (Structure theorem)**.** *Let $F = [m, M]$ be a fiber of the boundary map on $\mathrm{DM}_n$ and $D = M \setminus m$. Then $D$ is an antichain of diagonal points, and*

$$2^D \longrightarrow F, \qquad S \mapsto m \cup S$$

*is a lattice isomorphism; in particular $F$ is a Boolean lattice, with complementation $g \mapsto m \vee (M \wedge g^c)$. Conversely, for every antichain $D_0$ of diagonal points there is a fiber with $D = D_0$. Consequently*

$$\max_{\textit{fibers of } \mathrm{DM}_n} |F| \;=\; 2^{\binom{n}{\lfloor n/2 \rfloor}}.$$

*Proof.* $D$ is a diagonal antichain by Lemmas 3.2–3.3. For $S \subseteq D$, the set $m \cup S$ is an up-set: a point above some $\alpha \in S$ is either $\alpha$ itself or lies above a face point $\gamma > \alpha$ (Lemma 2.1 if diagonal; itself if a face point, since $\alpha$'s strict upper covers in $Q_n$ raise one coordinate, breaking a $(0,0)$ pair to $(0,1)$ or $(1,0)$), and such $\gamma \in M$, being a face point, lies in $m$; up-closure of $m$ finishes. Since $S$ avoids all face subcubes, $\partial(m \cup S) = \partial m$, so $m \cup S \in F$; conversely each $g \in F$ equals $m \cup (g \backslash m)$ with $g \backslash m \subseteq D$. The correspondence preserves unions and intersections, hence is a lattice isomorphism onto $F$; Boolean complementation transports to the stated formula.

Realizability: given a diagonal antichain $D_0$, let $m_0$ be the up-closure of the set of face points lying strictly above some point of $D_0$, and $M_0 := m_0 \cup D_0$. As above $M_0$ is an up-set, no point of $D_0$ lies in $m_0$ (a face point above $\alpha \in D_0$ is strictly above it, and points of $D_0$ are pairwise incomparable, so up-closure cannot reach $\alpha$; formally $\alpha \in m_0$ would give a face point $\gamma$ with $\gamma \leq \alpha$ — impossible since the generators of $m_0$ are $>$ some element of the antichain and $\alpha$ is minimal among $\geq$-relations within $M_0$), and $\partial m_0 = \partial M_0$ since $D_0$ avoids the faces. Hence the fiber of $\partial m_0$ has difference exactly $D_0$.

The maximum: diagonal points form a copy of $\{0,1\}^n$, whose maximal antichains have $\binom{n}{\lfloor n/2 \rfloor}$ elements by Sperner's theorem [14]; both bound and attainment follow. ☐

**Remark 3.5.** The proof of realizability is constructive and cheap to check by machine; at $n = 4$, the middle layer of the diagonal cube is an antichain of size $\binom{4}{2} = 6$ and the construction produces a fiber isomorphic to $2^6$ with all 64 members sharing their boundary — see §6.

**Corollary 3.6.** *All fiber cardinalities in* $\mathrm{DM}_n$ *are powers of two, at most* $2^{\binom{n}{\lfloor n/2 \rfloor}}$*; and* $M(2n) = \sum_{\textit{fibers}} 2^{|D|}$.

## 4. Every compatible sphere fills

**Definition 4.1.** A *sphere* in $\mathrm{DM}_n$ is a family $g = (g_i^e)_{i \leq n, e \in \{0,1\}}$, $g_i^e \in \mathrm{DM}_{n-1}$, satisfying the compatibility conditions $\partial_j^\delta g_i^\varepsilon = \partial_i^\varepsilon g_j^\delta$ (for $i \neq j$, indices adjusted as usual).

**Theorem 4.2.** *Every sphere is the boundary of an element of* $\mathrm{DM}_n$*; a canonical filler is the up-closure $m_g$ of the union of the prescribed* 1*-points, which is also the least filler.*

*Proof.* Identify each $g_i^e$ with its up-set inside the face subcube $Q_n^{(i,e)} \subseteq Q_n$, and let $m_g := \uparrow\!\left(\bigcup_{i,e} g_i^e\right)$. We must show $\partial_i^e m_g = g_i^e$. The inclusion $\supseteq$ is immediate. For $\subseteq$, let $\gamma \in Q_n^{(i,e)}$ lie in $m_g$, say $\gamma \geq \delta$ with $\delta \in g_j^{e'}$. If $(j,e') = (i,e)$, monotonicity of $g_i^e$ gives $\gamma \in g_i^e$. Otherwise let $\delta'$ agree with $\delta$ except that its $i$-th pair is set to the value prescribed by the face $(i,e)$; from $\delta \leq \gamma$ and $\gamma \in Q_n^{(i,e)}$ one checks $\delta \leq \delta' \leq \gamma$ componentwise, and $\delta'$ lies in both face subcubes $(i,e)$ and $(j,e')$. Monotonicity gives $\delta' \in g_j^{e'}$, compatibility transports this to $\delta' \in g_i^e$ (both sides restrict to the same element of $\mathrm{DM}_{n-2}$ on the common subcube), and monotonicity again gives $\gamma \in g_i^e$. Minimality is clear: any filler's up-set must contain every prescribed 1-point. $\square$

**Theorem 4.3** (Classification of boundaries)**.** *The assignments $g \mapsto m_g$ and $\varphi \mapsto \partial\varphi$ are mutually inverse bijections between spheres and the set of elements of $\mathrm{DM}_n$ all of whose minimal points are face points. Consequently spheres (equivalently, realizable boundaries) are in bijection with antichains of face points of $Q_n$, and*

$$B(n) \;=\; \#\{\textit{antichains of the poset of face points of } Q_n\},$$

*a Dedekind-type quantity for the $(4^n - 2^n)$-point poset of face points.*

*Proof.* A least fiber element $m$ contains no removable point, i.e. no minimal point that is diagonal (removing a minimal diagonal point preserves up-closure and all faces); conversely if every minimal point of $\varphi$ is a face point then no proper boundary-preserving shrinking exists, and Theorem 4.2 shows every sphere arises. An up-set is determined by its antichain of minimal points; those with all minimal points facial correspond exactly to antichains inside the subposet of face points. $\square$

**Remark 4.4.** For $n = 1$ the face points are the two incomparable points $(0,1), (1,0)$, giving $B(1) = 4$; for $n = 2$ the twelve face points give $B(2) = 82$; the machine count of antichains at $n = 3$ (a 56-point poset) confirms $B(3) = 4{,}829{,}136$, agreeing with the direct census (§6).

## 5. Median contraction of a fiber

The Boolean structure of Theorem 3.4 is a statement *about* the fiber; cubical applications additionally want paths *inside the free algebra* connecting two fillers. One fresh generator suffices.

**Theorem 5.1.** *Let $\varphi, \varphi' \in \mathrm{DM}_n$ have equal boundaries, and let $k$ be a fresh generator. Then*

$$\chi \;:=\; (\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi') \;\in\; \mathrm{DM}_{n+1}$$

*satisfies*

(1) $\partial_k^0 \chi = \varphi$ *and* $\partial_k^1 \chi = \varphi'$*;*
(2) *for every face $(i,e)$ of the original generators, $\partial_i^e \chi$ is the image of the common face $\partial_i^e \varphi = \partial_i^e \varphi'$ under the inclusion $\mathrm{DM}_{n-1} \hookrightarrow \mathrm{DM}_n$ adding $k$ — i.e. $\chi$ is constant in $k$ on the entire boundary.*

*Proof.* (1) Substituting $k := 0$ gives $\varphi \vee (\varphi \wedge \varphi') = \varphi$; $k := 1$ gives $\varphi' \vee (\varphi \wedge \varphi') = \varphi'$ (absorption). (2) Writing $\rho := \partial_i^e \varphi = \partial_i^e \varphi'$, the face of $\chi$ is $(\rho \wedge \neg k) \vee (k \wedge \rho) \vee \rho = \rho$, again by absorption. □

**Remark 5.2.** The shape of $\chi$ is a directed variant of the Birkhoff–Kiss median $(a \wedge b) \vee (b \wedge c) \vee (a \wedge c)$ [3]; the third term is essential — the naive interpolation $(\varphi \wedge \neg k) \vee (k \wedge \varphi')$ fails (2) precisely because the free De Morgan algebra does not satisfy the excluded middle for $k$. It also exhibits each fiber as a median-closed set: for $\varphi, \varphi', \psi$ in a fiber, the median of the three lies in the fiber (immediate from Lemma 3.1, and an instance of majority-closure phenomena classical in the theory of median algebras [2]). In cubical type theory, (1)–(2) say that $\chi$ is a homotopy rel boundary between the strict fillers $\varphi, \varphi'$ that is itself strict; iterating in fresh generators contracts the whole fiber. This is the lattice content of machine-checked results in [13].

## 6. The census

All counts below come from exhaustive enumeration in the representation by monotone functions ($n \leq 3$; at $n = 3$ this is the full algebra of $M(6) = 7{,}828{,}354$ elements, with boundaries grouped by their 6-tuples of faces), reproducible from the scripts accompanying the artifact; the structure-theorem invariants were verified independently: at $n = 2, 3$ every fiber's cardinality is a power of two, and the Boolean complement formula of Theorem 3.4 succeeded in all 2,130,934 non-singleton fibers at $n = 3$, with no failures.

| $n$ | $\lvert\mathrm{DM}_n\rvert = M(2n)$ | $B(n)$ | fiber sizes (size: count) | max |
|---|---|---|---|---|
| 1 | 6 | 4 | 1:2, 2:2 | 2 |
| 2 | 168 | 82 | 1:28, 2:38, 4:16 | 4 |
| 3 | 7,828,354 | 4,829,136 | 1:2698202, 2:1747364, 4:358284, 8:25286 | 8 |
| 4 | $M(8)$ | — | (not enumerated; see below) | 64 |

At $n \leq 3$ the maximum is $2^{\binom{n}{\lfloor n/2 \rfloor}} = 2, 4, 8$; at $n = 4$ full enumeration is out of reach ($M(8) \approx 5.6 \times 10^{10}$), but the realizability construction of Theorem 3.4 applied to the middle layer of the diagonal cube (an antichain of size $\binom{4}{2} = 6$) produces an explicit fiber isomorphic to $2^6$, all of whose 64 members were verified by machine to share their boundary; with the Sperner upper bound this settles max = 64 at $n = 4$. The column $B(3)$ was computed twice, independently: once by direct fiber-grouping, and once as the antichain count of Theorem 4.3.

The identity of Corollary 3.6 checks: $28 + 2 \cdot 38 + 4 \cdot 16 = 168$ and $2698202 + 2 \cdot 1747364 + 4 \cdot 358284 + 8 \cdot 25286 = 7828354$.

None of the sequences

$$B(n) = 4, 82, 4829136, \ldots \qquad \text{(nor the columns } 2, 28, 2698202;\ 2, 16, 25286\text{)}$$

matched any entry of the OEIS [9] at the time of writing (July 2026); we intend to contribute them. Interval-size statistics for the ambient monotone-function lattices, without the boundary grouping, were computed by Pawelski [11].

**Question 6.1.** Is there a closed form or efficient recursion for $B(n)$, i.e. for the number of antichains of the face-point poset of $Q_n$? The poset is a $2n$-cube minus its diagonal $n$-cube; its antichain count interpolates between Dedekind-type growth and the structure imposed by the excised diagonal.

**Question 6.2.** What is the distribution of $|D|$ over fibers as $n \to \infty$ — equivalently, of the number of "free diagonal points" of a random boundary? At $n = 3$ about 34.5% of elements lie in non-singleton fibers.

## 7. Outlook

Two directions seem worth recording. First, the results transfer verbatim to *sublanguages* of the interval theory: for the free bounded distributive lattice on $n$ generators (the "Dedekind" interval theory, no involution) every point of the evaluation cube is seen by some face, the difference $D$ is always empty, and fibers are singletons — boundary determines filler. The Boolean fibers are thus a specifically De Morgan phenomenon, quantifying the slack introduced by the involution; it would be interesting to treat the Kleene case (adding $x \wedge \neg x \le y \vee \neg y$), where the diagonal cube is constrained but not excised. Second, in the cubical-presheaf reading, Theorems 3.4, 4.2 and 5.1 equip the "strict layer" of cells over the De Morgan cube category with canonical fillers, Boolean fiber decompositions, and strict cylinders; whether this organizes into an algebraic weak factorization system with the strict layer as a cofibrant skeleton — a substitute for the Eilenberg–Zilber structure that the De Morgan cube category lacks — is the subject of ongoing work.

Artifact. The enumeration scripts, and the machine-checked type-theoretic counterparts of Theorems 3.4 and 5.1 (in a self-contained cubical proof kernel), are available in the repository `https://github.com/r-shibusawa/cubical-strict-j`, archived at DOI 10.5281/zenodo.21405961.

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